ERA5-Land Climate Data Note

August 2026

1 Purpose

This note documents the ERA5-Land data used for the rainfall-mortality work, including:

- what was downloaded
- why the downloads were regional
- where the daily files are stored
- which variables they contain; and
- which region folders should be used

The region names are confusing, and I should have cleaned up the folders before the hydrology processing began so that only the final region domains remained.

2 Why the Pull Was Regional

ERA5-Land is a global hourly dataset on a 0.1-degree grid. The full hourly archive is roughly 710 TB (8.3 TB for daily). Since the mortality data only cover specific countries and years, downloading the full global archive would have required processing a large amount of data we did not need.

I therefore downloaded regional bounding boxes for the countries and hydrology areas in the sample. The first covered India and the United States, including the relevant HydroBASINS intersections. I later added more regional downloads and a few small patches where the original bounding boxes did not fully cover the countries in our sample.

3 Downloads and Daily Aggregation

I first downloaded the data through the Copernicus CDS API. Large region-month requests were slow and often sat in the queue. I used multiple API keys, but could still run only a limited number of jobs at once.

I later moved most downloads to the DestinE Earth Data Hub, which was much faster and allowed more region-month jobs to run in parallel. The processing steps were otherwise the same:

1. download hourly ERA5-Land data for one region-month
2. aggregate the hourly data to daily variables
3. save the daily NetCDF files
4. delete the hourly files, which are too large to keep

Precipitation requires an extra step because ERA5-Land total precipitation is accumulated rather than reported as an hourly increment. Timestamps are shifted back by one hour so that the value recorded at midnight, which contains the final total for the previous day, is assigned to that day. The daily maximum is then taken and converted from meters to millimeters.

4 Daily Files and Variables

The daily files are stored here:

`/project/cil/gcp/climate/ERA5-LAND/raw/daily/<region>/<variable>/`

Files follow this naming pattern:

`era5land_<region>*<variable>*<year>.nc4`

Variable	Description	Units	Daily calculation
`tmin`	Minimum 2m air temperature	°C	Minimum hourly temperature, converted from K to °C
`tmean`	Mean 2m air temperature	°C	Mean hourly temperature, converted from K to °C
`tmax`	Maximum 2m air temperature	°C	Maximum hourly temperature, converted from K to °C
`rh_mean`	Mean relative humidity	%	Calculated hourly from temperature and dew point, then averaged
`q_mean`	Mean specific humidity	g/kg	Calculated hourly from dew point and surface pressure, then averaged
`swe_mean`	Mean snow water equivalent	mm	Converted from m to mm, then averaged
`tp_total`	Total precipitation	mm	Timestamps shifted back one hour, followed by the daily maximum (last hour is the accumulated total, therefore the max) and conversion from m to mm
`wind_max`	Maximum 10m wind speed	m/s	Wind speed calculated hourly from $u10$ and $v10$, followed by the daily maximum

5 Region Folders

The regional boxes were created for different parts of the sample and later expanded where needed. The original USA box included the relevant HydroBASINS areas and covered most of Mexico, but not southern Mexico. I therefore downloaded `mexico_south` and merged it with `usa`.

The India box also included the relevant HydroBASINS areas. I later downloaded `asia_east` to cover other Asian countries in the sample and merged it with `india`. The original `latam` box was intended to cover all of Latin America but missed a small part of eastern Brazil. This caused problems during the Brazil aggregation, so I downloaded `brazil_east` to fill the gap.

The result is three final merged folders whose names do not reflect their full coverage.

Table 1: Final merged regions

Current folder	Rename to	Combined from	Reason for additional box
`asia_combined`	`asia`	`india` and `asia_east`	To cover other Asian countries in the sample
`mexico`	`north_america`	`usa` and `mexico_south`	The original box did not cover southern Mexico
`brazil`	`latin_america`	`latam` and `brazil_east`	The original box missed a small part of eastern Brazil

These folders already contain the final merged daily files, they just bneed to be renamed properly. The `europe` and `australia` folders can keep their current names.

The following component folders should be moved into a separate `components` or `archive` folder:

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- `india`
- `asia_east`
- `usa`
- `mexico_south`
- `latam`
- `brazil_east`

The main daily data directory should eventually contain only the final region folders:

- `asia`
- `north_america`
- `latin_america`

- `europe`
- `australia`

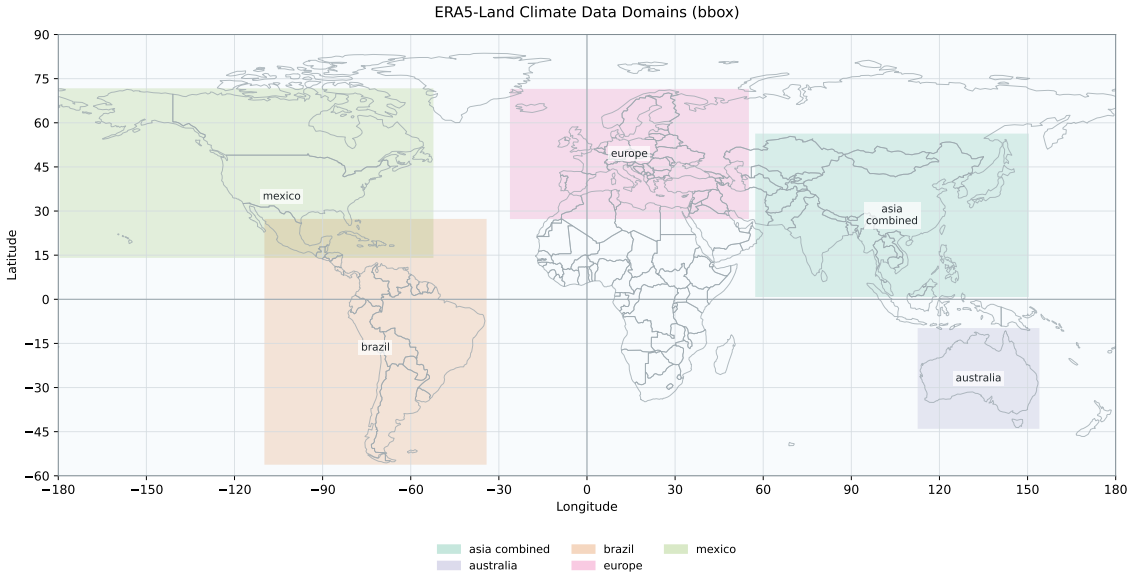


Figure 1: Final ERA5-Land climate data domains.

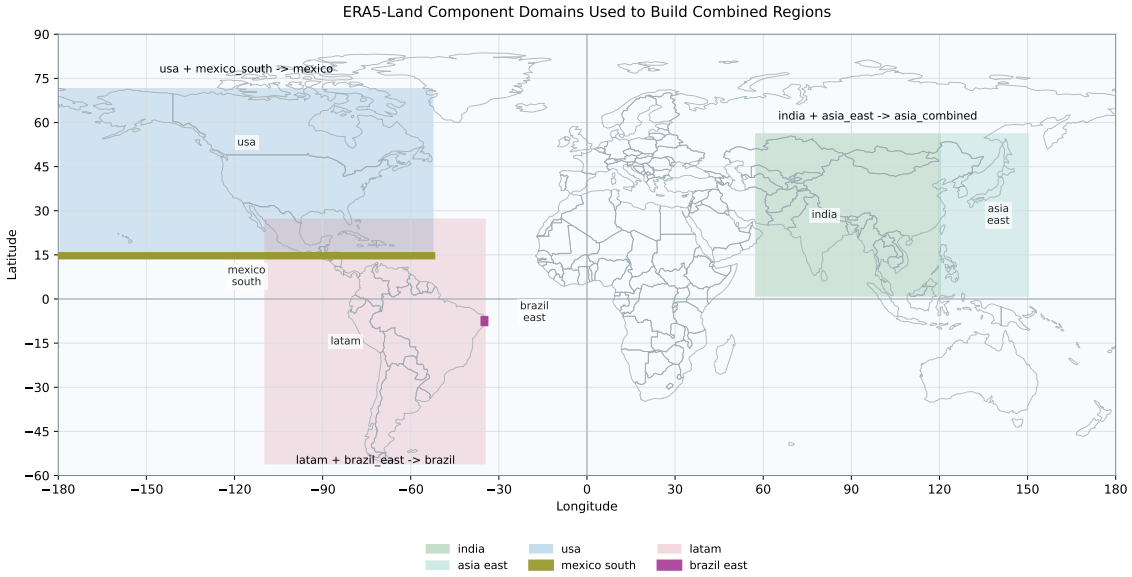


Figure 2: Component domains used to build merged domains.

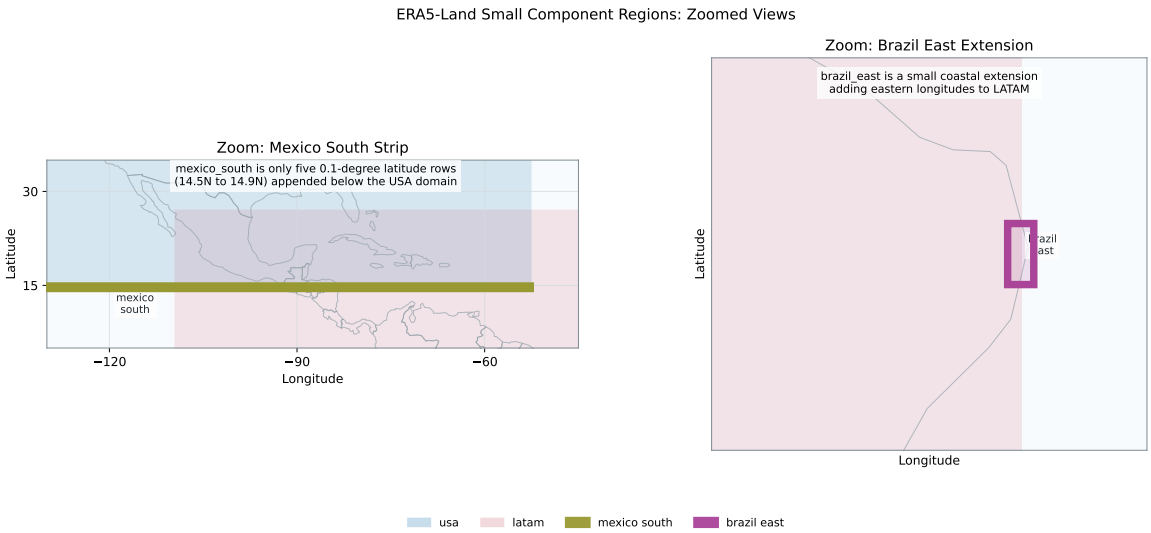


Figure 3: Zoomed views of the small Mexico and Brazil patches.

6 Missing Values and Precip Checks

Because ERA5-Land covers land, cells over oceans and other water bodies are often missing within the rectangular bounding boxes. Some regions therefore have high missing shares, especially where the boxes include large ocean areas.

I checked all 460 daily precip files (across regions and years) in

`/project/cil/gcp/climate/ERA5-LAND/raw/daily/<region>/tp_total/.nc4`

and found no negative values.

Monthly `tp` in `/project/cil/gcp/climate/ERA5-LAND/processed/monthly` was also non-negative.

Negative values appeared only in the derived variables `rainfall` and `rainfall_flow`.

7 Key Paths

```
Daily climate files:
/project/cil/gcp/climate/ERA5-LAND/raw/daily

Monthly processed outputs:
/project/cil/gcp/climate/ERA5-LAND/processed/monthly
```